NATIONAL INCOME ANALYSIS

1. MEANING

National Income is a measure of the money value of goods and services becoming available to a nation from economic activities. It can also be defined as the total money value of all final goods and services produced by the nationals of a country during some specific period of time – usually a year – and to the total of all incomes earned over the same period of time by the nationals.

2. DIFFERENT CONCEPTS OF NATIONAL INCOME

Gross Domestic Product
The money value of all goods and services produced within the country but excluding net income from abroad.

Gross National Product
The sum of the values of all final goods and services produced by the nationals or citizens of a country during the year, both within and outside the country.

Net National Product
The money value of the total volume of production (that is, the gross national product) after allowance has been made for depreciation (capital consumption allowance).

Nominal Gross National Product
The value, at current market prices, of all final goods and services produced within some period by a nation without any deduction for depreciation of capital goods.

Real Gross National Product
This is the national output valued at the prices during some base year or nominal GNP corrected for inflation.

3. NATIONAL INCOME ACCOUNTING

This refers to the measuring of the total flow of output (goods and services) and of the total flow of inputs (factors of production) that pass through all of the markets in the economy during the same period.

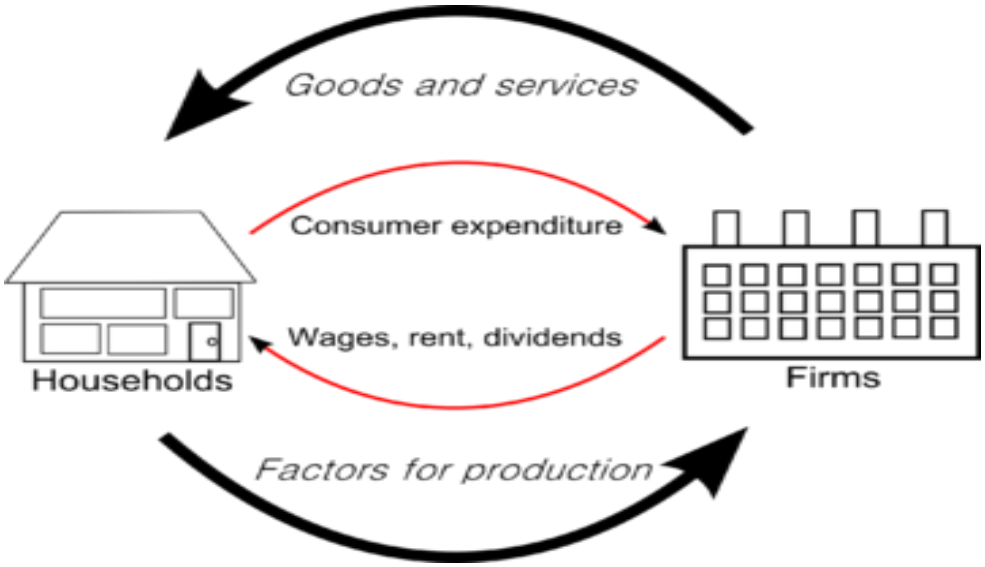
To see exactly what national income includes, how it is measured, and what it can tell us, we start with economic models: By economic models we mean:

A simplification of a real world or a practical situation aimed at explaining that situation within a set of assumptions.

The Circular Flow of Income and Expenditure

This is an economic model illustrating the flow of payments and receipts between domestic firms and

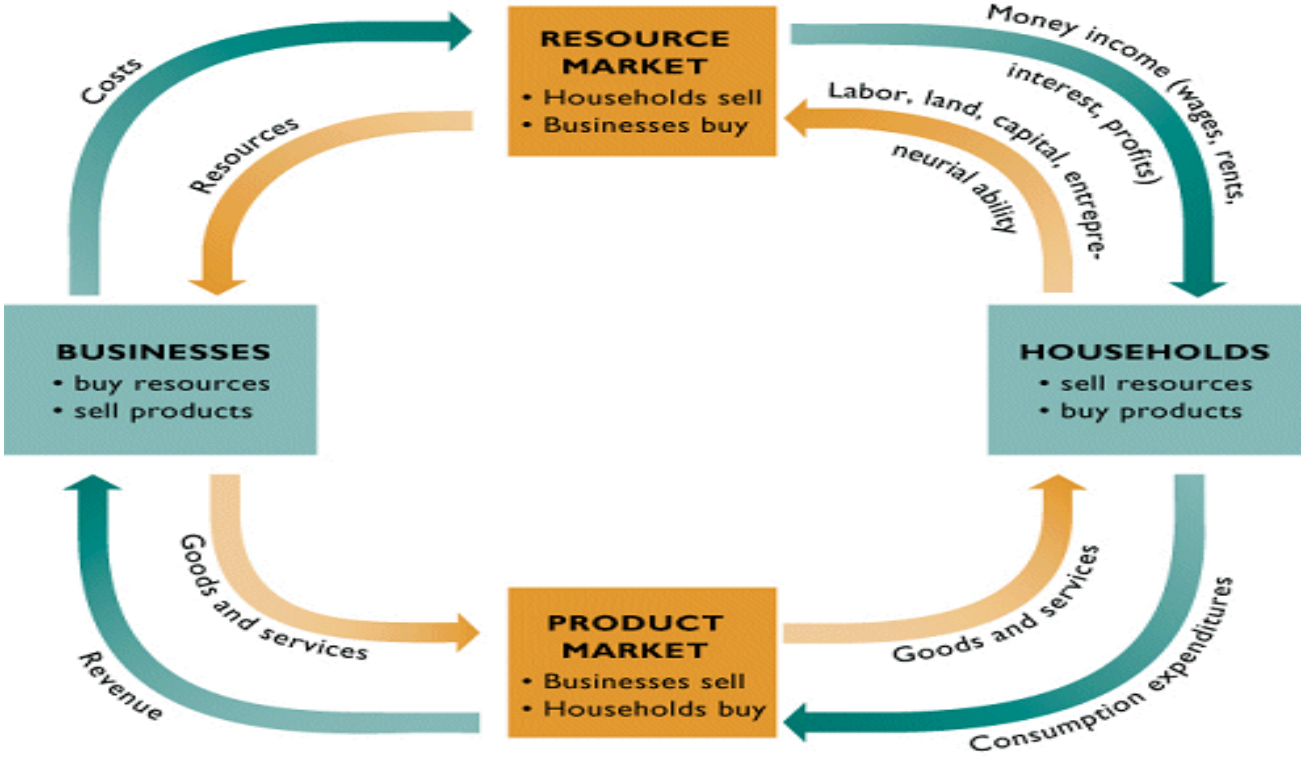
domestic households. The households supply **factor services** to the firms. In return, they get **factor incomes**. With factor incomes, they buy **goods and services** from the firms. These flows can be illustrated diagrammatically as follows for a two sector economy:



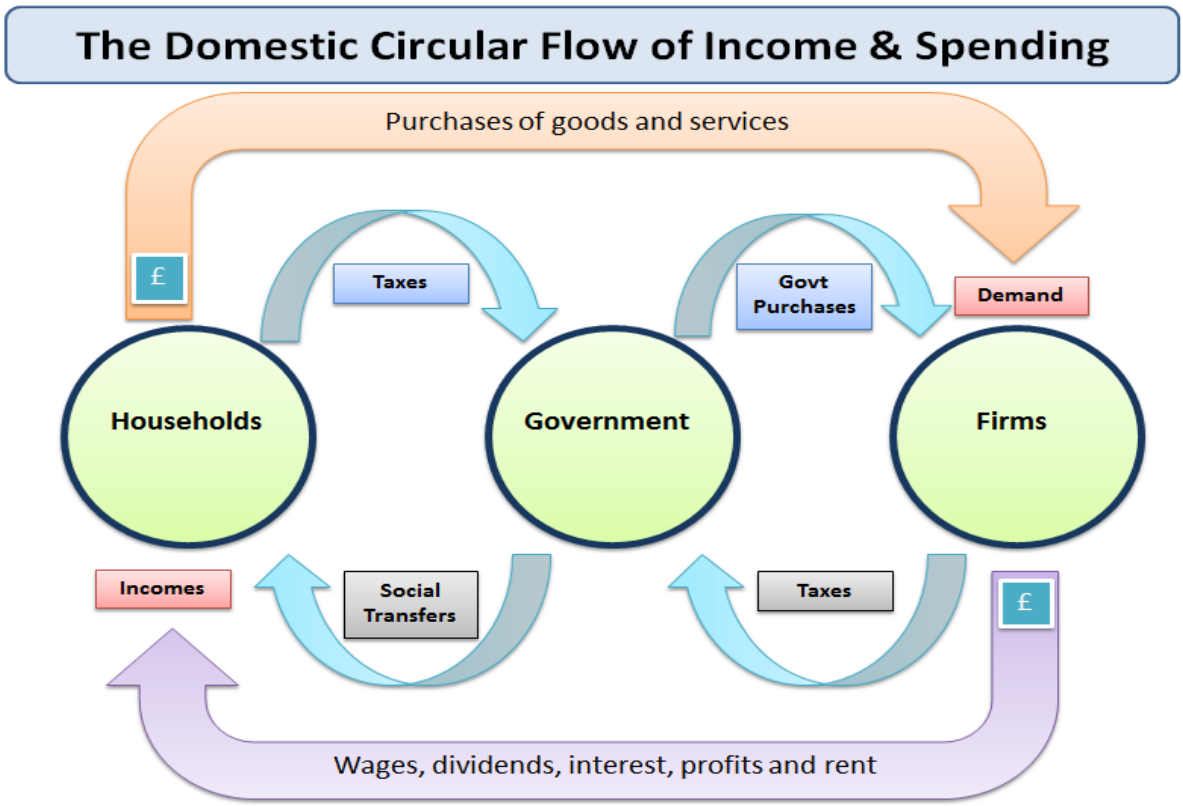
Two Sector economy = Household + Firms

Y = C + I

The points at which flows from one sector meets the other sector and generate other flows are called **critical points**. In the above diagram, the critical points are A, B and C. At A, the flow of factor services from the households sector meets the firm sector and generates the **flow of factors incomes** from the firms to the households. At B, the flow of factor incomes meets the household sector and generates the flow of **consumer spending**. At C, the flow of consumer spending meets the firms sector and generates the flow of **goods and services**.


RESOURCE MARKET
• Households sell
• Businesses buy
Money income (wages, rents, interest, profits)
Labor, land, capital, entrepre-neurial ability
Costs
Resources
BUSINESSES
• buy resources
• sell products
HOUSEHOLDS
• sell resources
• buy products
Goods and services
Revenue
PRODUCT MARKET
• Businesses sell
• Households buy
Goods and services
Consumption expenditures

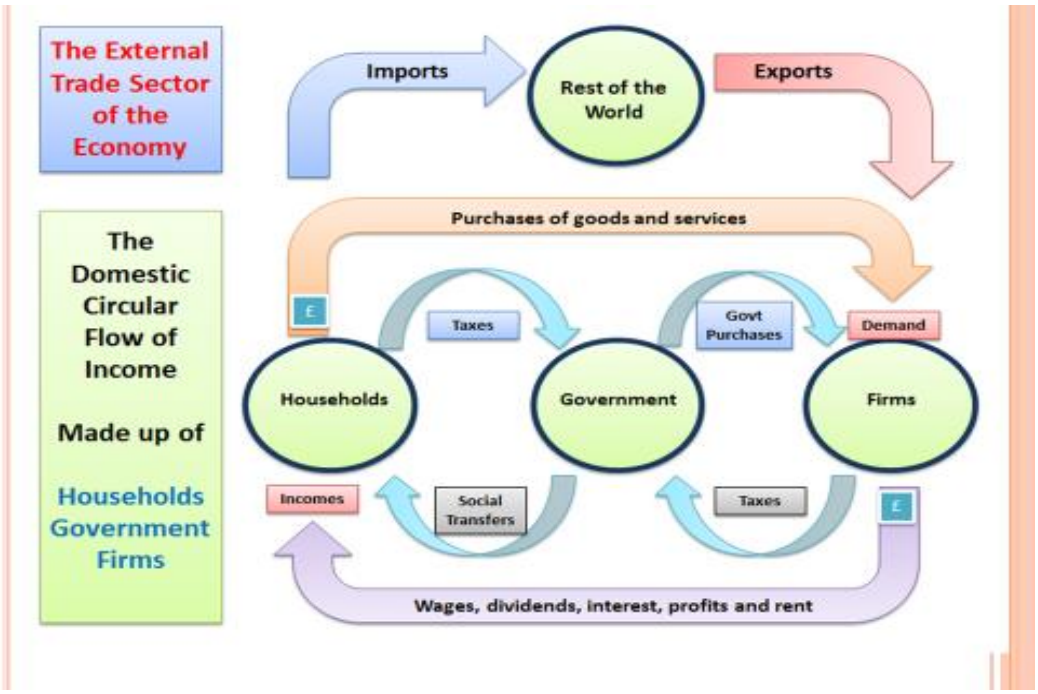
CIRCULAR FLOW OF INCOME FOR A THREE SECTOR ECONOMY



Three Sector economy = Household + Firms + Governemt

Y = C + I + G

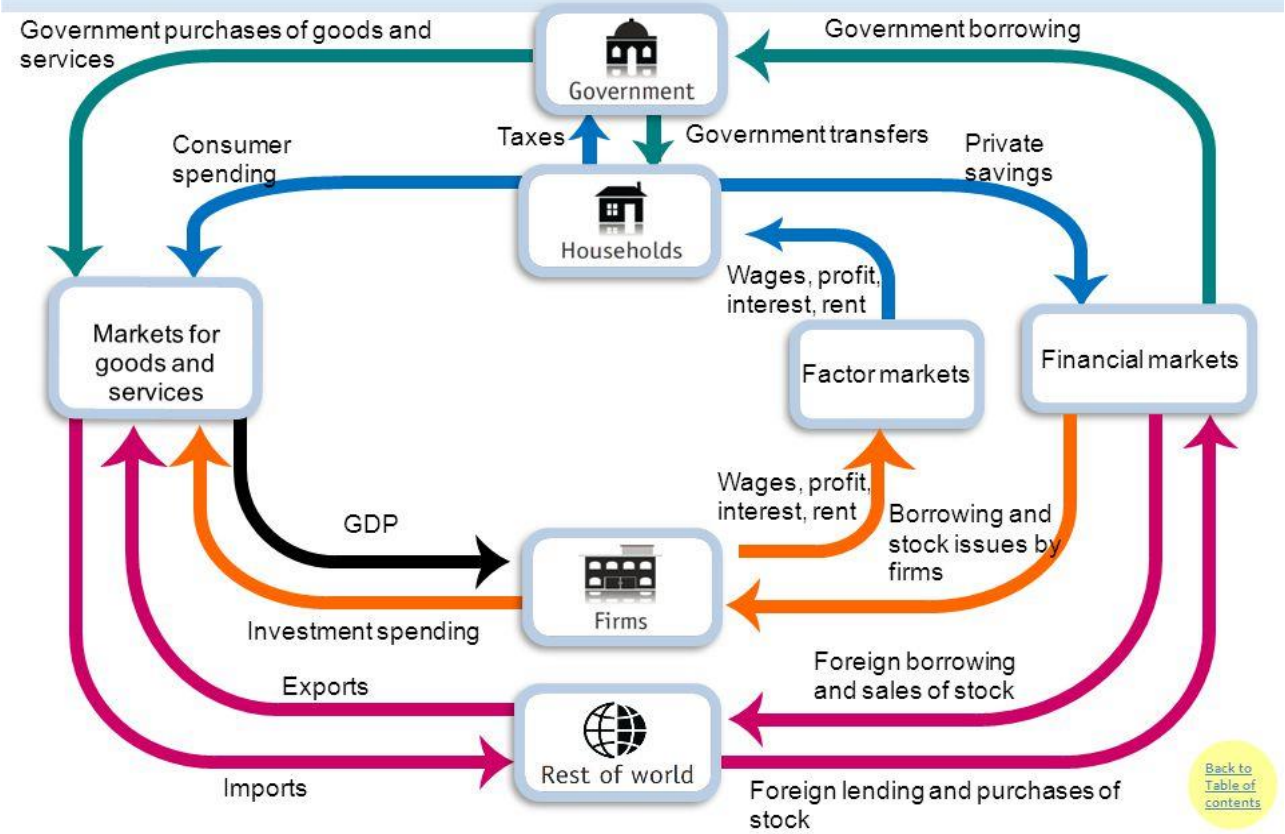
CIRCULAR FLOW OF INCOME FOR A FOUR SECTOR ECONOMY



Four Sector economy = Household + Firms + Government + Exports - Imports

Y = C + I + G + X - I

An Expanded Circular-Flow Diagram



4. APPROACHES TO MEASURING NATIONAL INCOME

The compilation of national income statistics is a very laborious task. The total wealth of a nation has to be added up and there are millions of nationals. Moreover, in order to double check and triple check the statistics, the national income statistician has to work out the figures out in three different ways, each way being based on a different aspect. The three aspects are:

a. **The national output: -** The creation of wealth by the nation's industries. This is valued at factor cost, so it must be the same as b) below.

b. **The national income**: - The incomes of all the citizens.

c. **The national expenditure** because whatever we receive we spend, or lend to the banks to invest it, so that the addition of all the expenditure should come to the same as the other two figures.

Put in its simplest form we can express this as an identity:

National output ≡ National Income ≡ National Expenditure.

(i) Using Total Expenditure for Calculating National Income

The expenditure approach centres on the components of ***final demand*** which generate production. It thus measures GDP as the total sum of expenditure on ***final goods and services*** produced in an economy. It includes

- All consumers' expenditure on goods and services, except for the purchase of new houses which is included in gross ***fixed capital formulation***.
- All general ***government final consumption***. (all current expenditure by ***central and local*** government on goods and services, including wages and salaries of government employees.)
- ***gross fixed capital formation or expenditure on fixed assets*** (buildings, machinery, vehicles etc) either for replacing or adding to the stock of existing fixed assets. This is the major part of the investment which takes place in the economy.
- ***value of physical increases in the stocks, or inventories,*** during the course of the year. The total of all this gives us ***Total domestic expenditure*** (TDE).
- expenditure on exports to the TDE and arrive at a measure known as ***Total Final Expenditure***. It is so called because it represents the total of all spending on final goods.
- the final expenditure is on ***imported goods*** and we therefore ***subtract spending on imports***.

Having done this we arrive at a measure known ***as gross domestic product at market prices***.

To gross domestic product at market price we

- ***subtract the taxes on expenditure*** levied by the government and add on the amount of subsidy.

When this has been done we arrive at ***Gross Domestic Product at factor cost.***

National Income however is affected ***by rent, profit interest*** and dividends paid to, or received from, overseas. This is added to GDP as net property income from abroad.

This figure may be either ***positive or negative.*** When this has been taken into account we arrive at the ***gross national product at factor cost.***

As production takes place, the capital stock of a country wears out. Part of the gross fixed capital formation is therefore, to replace worn out capital and is referred to as ***Capital Consumption***.

When this has been subtracted we arrive at a figure known as the ***net national product***. Thus, summarising the above, we can say:

$$Y = C + I + G + (X - M)$$

(ii) Using Factor Incomes for Calculating National Income

A second method is to **sum up all the incomes** to individuals in the form of **wages, rents, interests** and **profits** to get domestic incomes.

- This is because each time something is produced and sold someone obtains income from producing it.
- It follows that if we add up all incomes we should get the value of total expenditure, or output.
- Incomes earned for purposes other than rewards for producing goods and services are ignored.
- Such incomes are **gifts, unemployment or relief benefits, lottery, pensions, grants** for students etc.
- These payments are known as **transfer income (payments)** and including them will lead to **double counting.**
- The test for inclusion in the national income calculation is therefore that there should be a **"quid pro quo"** that the money should have been paid against the exchange of a good or service.
- Alternatively, we can say that there should be a **"real"** flow in the opposite direction to the money flow. We must also include income obtained from **subsistence output.**
- This is the opposite case from transfer payments since there is a flow of real goods and services, but no corresponding money flow.
- It becomes necessary to "impute" values for the income that would have been received. Similarly workers may, in addition to cash income, receive **income in kind**; if employees are provided with **rent free housing**, the rent which they would have to pay for those houses on the open market should, in principle, be "imputed" as part of their income from employment. The sum of these incomes gives **gross domestic product GDP**.
- This includes incomes earned by foreigners at home and excludes incomes earned by nationals abroad. Thus, to **Gross Domestic Income** we add **Net property Income** from abroad.
- This gives **Gross National Income.**
- From this we deduct **depreciation** to give **Net National Income.**

(iii) Using the National Output for Calculating National Income

A final method which is more direct is the **"output method"** or the **value added approach**.

- This involves adding up the total contributions made by the various sectors of the economy.
- "Value Added" is the **value added by each industry to the raw materials or processed products that it has bought from other industries before passing on the product to the next stage in the production process**.
- This approach therefore centres on **final products.**

- Final products will include **capital goods** as well as **consumer goods** since while **intermediate goods** are used up during the period in producing other goods, capital goods are not used up (apart from "wear and tear" or depreciation) during the period and may be thought of as consumer goods **"stored up"** for future periods. Final output will include **"subsistence output",** which is simply **the output produced and consumed by households themselves.**
- Because subsistence output is not sold in the market, some assumption has to be made to value them at some price.
- We also take into account the **final output of government,** which provides services such as education, medical care and general administrative services.
- However, since state education and other governmental services are not sold on the market we shall not have market prices at which to value them.
- The only obvious means of doing this is to value public services at what it costs the government to supply them that is, by the **wages bill spent on teachers, doctors, and the like.**
- When calculating the GDP in this matter it is necessary to avoid **double counting**.

5. SOME DIFFICULTIES IN MEASURING NATIONAL INCOME

National Income Accounting is beset with several difficulties. These are:

a. What goods and services to include

Although the general principle is to take into account only those products which change hands for money, the application of this principle involves some arbitrary decisions and distortions. For example, unpaid services such as those performed by a housewife are not included but the same services if provided by a paid housekeeper would be.

Many farmers regularly consume part of their produce with no money changing hands. An imputed value is usually assigned to this income. Many durable consumer goods render services over a period of time. It would be impossible to estimate this value and hence these goods are included when they are first bought and subsequent services ignored. Furthermore, there are a number of governmental services such as medical care and education, which are provided either 'free' or for a small charge. All these provide a service and are included in the national income at cost. Finally, there are many illegal activities, which are ordinary business and produce goods and services that are sold on the market and generate factor incomes.

b. Danger of Double Counting

The problem of double counting arises because of the inter-relationships between industries and sectors. Thus we find that the output of one sector is the input of another. If the values of the outputs of all the sectors were added, some would be added more than once, giving an erroneously large figure of national income. This may be avoided either by only including the value of the final product or alternatively by summing the values added at each stage which will give the same result.

Some incomes such as social security benefits are received without any corresponding contribution to production. These are **transfer payments** from the taxpayer to the recipient and are not included. **Taxes** and **subsidies** on goods will distort the true value of goods. To give the correct figure, the former should not be counted as an increase in national income for it does not represent any growth in real output.

c. Inadequate Information

The sources from which information is obtained are not designed specifically to enable national income to be calculated. Income tax returns are likely to err on the side of understatement. There are also some

incomes that have to be estimated. Also, some income is not recorded, as for example when a joiner, electrician or plumber does a job in his spare time for a friend or neighbour. Also information on foreign payments or receipts may not all be recorded.

Factors affecting the size of a National Income

The size of a nation's income depends upon the quantity and quality of the factor endowments at its disposal. A nation will be rich if its endowments of natural resources are large, its people are skilled, and it has a useful accumulation of capital assets. The following points are of interest:

a) Natural Resources

These include the minerals of the earth; the timber, shrubs and pasturage available; the agricultural potential (fertile soil, regular rainfall, temperature or tropical climate); the fauna and flora; the fish; crustacea etc of the rivers and sea; the energy resources, including oil, gas, hydro-electric, geothermal, wind and wave power.

b) Human Resources

A country is likely to prosper if it has a large population; literate and numerate sophisticated and knowledgeable about wealth creating processes. It should be well educated and skilled, with a nice mixture of theory and practice. It should show enterprise, being inventive, energetic and determined in the pursuit of a better standard of living.

c) Capital Resources

A nation must create and then conserve capital resources. This includes not only tools, plant and machinery, factories, mines, domestic dwellings, schools, colleges, etc, but a widespread infrastructure of roads, railways, airports and ports. Transport creates the utility of space. It makes remote resources accessible and high-cost goods into low-cost goods by opening up remote areas and bringing them into production.

d) Self-sufficiency

A nation cannot enjoy a large national income if its citizens are not mainly self-supporting. If the majority of the enterprises are foreign –owned there will be a withdrawal of wealth in the form of profits or goods transferred to the investing nation.

e) Political Stability

6. USES OF NATIONAL INCOME FIGURES

- We need national income statistics to **measure the size of the "National cake'** of goods and services available for competing uses of private consumers, government, capital formation and exports (less imports).

- National Income statistics are also used in **comparing the standard of living of a country over time**

- And also the **standards of living between countries**.

- National Income Statistics provide information on the **stability of performance of the economy over time** e.g. a steadily increasing income would be indicative of increasing national income.

- If National Income Statistics are disaggregated it would enable us to assess the **relative importance of the various sectors in the economy**. This is done by considering the contribution of the various sectors to Gross National Product over time. Such information is crucial for planning purposes for it reveals to planners where constraints to economic development lie. It therefore becomes possible to design a development strategy that eventually would overcome these problems. This central contribution could be in the form of employment or the production of goods and services.

- By assessing exports and imports as a percentage of Gross national Product i.e. using national statistics, it is possible to **determine the extent to which a country depends on external trade**.

- National Income Statistics also help in **estimating the saving potential and hence investment potential of a country.**

Real Vs Nominal GNP: "Deflating" by a price Index

One of the problems that confront economists when measuring GNP is that they have to use money as the measuring rod. These days however, inflation sends the general price level up and up clearly this means that the yardstick stretches in their hands every day.

Economists repair most of the damage wrought by the elastic yardstick by using a price index. The price index used to remove inflation (or "deflate" the GNP) is called the GNP deflator. The GNP deflator is defined as the ratio of nominal GNP to real GNP. It is constructed as follows:

$$\textbf{GNP Deflator} = \frac{\text{Nominal GNP}}{\text{Real GNP}}$$

$$\textbf{Real GNP} = \frac{\text{Value at current Price}}{\text{CPI}}$$

Where CPI is Consumer Price Index
The GNP deflator is useful because it includes prices on all goods and services in GNP.

PER CAPITA INCOME
By National Income is meant the value of outputs produced within a year. Income per capita is simply the National Income divided by the population of the country in a year.

$$\text{INCOME PER CAPITA} = \frac{\text{National Income}}{\text{Population}}$$

It **shows the standard of living a country can afford for its people**. The level of income per capita is determined by the size of a country's population. **The higher is the rate of growth of population; the lower is the rate of growth of income per capita.**

Per capita income is a theoretical rather than a factual concept. It shows what the share of each individual's National Income would be if all citizens were treated as equal. In a real world situation there exists considerable inequality in the distribution of income especially in the third world countries.

NATIONAL INCOME AND WELFARE

The relationship between National Income and Welfare is best explained in terms of economic growth (By economic growth is meant capacity expansion). The effect of economic growth is an increase in the National Income. This increase in National Income has several effects on a country's citizens.

1) Assuming a fair distribution of income, the average citizen would be in a position to enjoy a higher living standard.

2) The ordinary households or persons could be able to afford luxury commodities. NB: luxury differs in its definition from one country to another and the determining factor being the level of income. e.g. clothing can be a luxury for some people.

3) It enables the ordinary household to afford leisure which may be regarded as luxury i.e. reducing working hours.

Points 1, 2 and 3 are based on assumption that there exists a fair distribution of the National cake. This may not be the case in fact **it is disastrous to rely on GNP**, its growth rate and GNP per capita as indicators of economic well being.

GNP per capita e.g. gives no indication of how National Income is actually distributed and who is benefiting most from the growth of production. A rising level of absolute and per capita GNP may camouflage the fact that the poor are not better than before. In fact the calculation of GNP, and especially its rate of growth is in reality largely a calculation of the rate of growth, of the incomes of upper 20% who receive a disproportionately large share of the National Product. It is, therefore, unrealistic to use GNP growth rates as an index of improved economic welfare for the general public.

Example: Assuming a 10 people economy and assuming 9 of them had no income and the 10th person receives 100 units of income the GNP for this economy would be 100 units of income and per capita income would be 10 units. Suppose everyone's income increases by 20% so that GNP rises to 120 units per capita income would rise to 12 units. However for the 9 people without income before and currently such a rise in per capita income provides no cause for rejoicing since the one rich individual still has the income. In this case we observe that GNP instead of being a welfare index of a society as a whole is merely increasing the welfare of a single individual.

This exchange though an extreme case is indicative of what happens in the real life situation where incomes are very unequally distributed.

NATIONAL INCOME AND STANDARDS OF LIVING

Standard of living refers to the quantity of goods and services enjoyed by a person. These goods may be provided publicly, such as in the case of health care or education or they may be acquired by direct purchase. It also includes the less easily quantifiable aspects of living such as terms and conditions of employment and general living environment.

National Income figures can be used to measure the standard of living at a particular point of time and over time. This is done by working out the per capita income of the country. **By per capita income we mean: the value of goods and services received by the average man.** Per capita income is obtained by dividing the National Income by the Total population. If the per capita income is high, it can be deduced that the standard of living is high.

PROBLEMS OF USING PER CAPITA INCOME TO COMPARE STANDARD OF LIVING OVER TIME

1) **The composition of output may change**. e.g. more defence-related goods may be produced and less spent on social services, more producer goods may be made and less consumer goods, and there may be a surplus of exports over imports representing investment overseas. Standards of living depend on the quantity of consumer goods enjoyed.

2) **Over time prices will change.** The index of retail prices may be used to express the GNP in real terms but there are well known problems in the use of such methods.

3) **National Income may grow but this says nothing about the distribution of that income**. A small group may be much better off. Other groups may have a static standard of living or be worse off.

4) **Any increase in GNP per capita may be accompanied by a decline in the general quality of life.** Working conditions may have deteriorated. The environment may have suffered from various forms of pollution. These non-monetary aspects are not taken into account in the estimates of the GNP.

5) Finally the **national income increases when people pay for services which they previously carried out themselves**. If a housewife takes an office job and pays someone to do her housework, national income will increase to the extent of both persons' wages. Similarly a reduction in national income would occur if a man painted his house rather than paying a professional painter to do the same. Changes of the above type mean that changes in the GNP per capita will only imperfectly reflect changes in the standard of living.

PER CAPITA INCOME AND INTERNATIONAL COMPARISONS

Per capita income figures can also be used to compare the standards of living of different countries. Thus if the per capita income of one country is higher than that of another country, the living standard in the first country can be said to be higher. Such comparisons are made by aid giving international agencies like the United Nations and they indicate the relevant aid requirements of different countries.

But there are major problems in using **real income per head** (per capita income) to measure the standard of living in different countries. First there is the whole set of **statistical problems** and, secondly, there are a number of difficult **conceptual problems** or problems of interpretation.

i. **Inaccurate estimates of population:** The first statistical problem in calculating income per head particularly in less developed countries is that we do not have very accurate population figures with which to divide total income.

ii. **Specific items which are difficult to estimate:** Another data problem, as already mentioned, is that data for depreciation and for net factor income from abroad are generally unreliable. Hence although we should prefer figures for "the' national income, we are likely to fall back on GDP, which is much less meaningful figure for measuring income per head. Inventory investment and work-in-progress are also difficult items to calculate.

iii. **Non-marketed subsistence output and output of government:** some output like subsistence farming and output of government are not sold in the market. These are measured by taking the cost of the inputs. In one country, however, salary of doctors for instance, might be higher and their quality low compared to

another country. Although the medical wage bill will be high, the "real consumption" of medical care in the former might be lower. Since "public consumption" is an important element in national income, this could affect comparisons considerably.

Also in making international comparisons it is assumed that the complied national income figures of the countries being compared are equally accurate. This is not necessarily the case. If, for example, in one country there is a large subsistence sector, a lot of estimates have to be made for self-provided commodities. The national figures of such a country will, therefore, be less accurate than those of a country whose economy is largely monetary or cash economy.

iv. **Different degrees of income distribution**: If the income of one country is evenly distributed, the per capita income of such a country may be higher than that of another country with a more evenly distributed income, but this does not necessarily mean that most of its people are at a higher living standard.

v. **Different Types of Production:** If one country devotes a large proportion of its resources in producing non-consumer goods e.g. military hardware, its per capita income may be higher than that of another country producing largely consumer goods, but the standard of living of its people will not necessarily be higher.

vi. **Different forms of Published National Income figures**: The per capita income figures used in international comparisons are calculated using the published figures of national income and population by each country. For meaningful comparisons, both sets of national income figures should be in the same form i.e. both in real terms or both in money terms, the latter may give higher per capita income figures due to inflation, and thus give the wrong picture of a higher living standard. On the other hand, if both sets are in money terms the countries being compared should have the same level of inflation. In practice, this is not necessarily the case.

vii. **Exchange Rates:** Every country records its national income figures in its own currency. To make international comparisons, therefore, the national income figures of different countries must have been converted into one uniform currency. Using the official exchange rates does this. Strictly speaking, these apply to internationally traded commodities, which normally form a small proportion of the national production. The difficulty is that these values may not be equivalent in terms of the goods they buy in their respective commodities i.e. the purchasing power of the currencies may not be the same as those reflected in the exchange rate.

viii. **Difference in Price Structures:** Differences in the relative prices of different kinds of goods, due to differences in their availability, mean that people can increase their welfare if they are willing to alter their consumption in the direction of cheaper goods. The people in poor countries probably are not nearly as badly off as national income statistics would suggest, because the basic foodstuffs, which form an important part of their total consumption, are actually priced very low.

ix. **Income in relation to Effort:** The first conceptual problem in calculating income per head is to look at goods and services produced in relation to the human effort that has gone into producing them. Obviously if people work harder, they will be able to get more goods; but they may prefer the extra leisure. Indeed, the amount of leisure that people want depends in part on their level of income. Strictly, therefore, we should take income per unit of labour applied. It is largely because this would be statistically awkward that economists prefer to take real income per head.

x. **Differences in size:** A problem which is both conceptual and statistical is due to the transport factor. If two countries are of different sizes, the large country may devote a large proportion of its resources in

developing transport and communication facilities to connect the different parts of the country. This will be reflected in its national income, but the standard of living of its people will not necessarily be higher than that of smaller country, which does not need these facilities to the same extent.

xi. **Differences in Taste:** Another formidable difficulty is that tastes are not the same in all countries. Also in different countries the society and the culture may be completely different thus complicating comparisons of material welfare in two countries. Expensive tastes are to some extent artificial and their absence in poor countries need not mean a corresponding lack of welfare. Tastes also differ as regards the emphasis on leisure as against the employment of the fruit of labour: if in some societies people prefer leisure and contemplation, who is to say this reduces their welfare as compared to those involved in the hurly-burly of life and labour in modern industry?

xii. **Different climatic zones:** If one country is in a cold climate, it will devote a substantial proportion of its resources to providing warming facilities, e.g. warm clothing and central heating. These will be reflected in its national income, but this does not necessarily mean that its people are better off than those in a country with a warm climate.

xiii. **Income per head as index of economic welfare:** We cannot measure material welfare on an arithmetic scale in the same way as we measure real income per head. For instance, if per capita income increases, material welfare will increase; but we cannot say by how much it has increased, and certainly that it has increased in proportion.